

Research on path planning method of forging handling robot based on combined strategy

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Abstract—In the robot navigation module, autonomous path planning is at the core, and the path planning control strategy is the key factor. Traditional robot path planning algorithms generally have some inherent defects, and they cannot seamlessly integrate local and global path planning methods, resulting in low efficiency. This paper proposes a combination strategy of local and global path planning algorithms. In the global algorithm, the potential field ant colony algorithm is adopted. Before the robot starts, it will plan a relatively optimal path on the specified electronic map, which solves the problems of premature convergence and low efficiency of the traditional ant colony algorithm; in terms of local algorithms The improved artificial potential field method is adopted, which carries out real-time path planning for the unknown environment, and solves the problems of local minima and unreachable targets in the traditional potential field method. In this paper, the proposed improved method is simulated and analyzed, and the results show that the proposed control method is feasible.

Keywords—Transport Robot, Path Planning, Combination Strategy, Ant Colony Algorithm

I. INTRODUCTION

The forging industry occupies an important position in the manufacturing industry. The harsh environment such as vibration, noise, high temperature, smoke and dust in the forging workshop is harmful to the health of workers. In order to improve the production efficiency, operation conditions and working environment, under the concept of "green forging", the use of forgings handling robot for autonomous handling can reduce labor costs and avoid damage to people caused by the factory environment. An effective and feasible autonomous path planning control strategy is a key indicator of the degree of autonomy and intelligence of the handling robot^[1]. Therefore, the study of path planning control strategy has received extensive attention from foreign scholars. Barraquand et al.^[2] proposed to gradually build a graph that connects the local minima of the potential field defined in the robot configuration space, and search the graph at the same time until the target configuration is obtained. C.Rahul Kala et al.^[3] used memory-based search for local optimization, in which the best path between different intersections in the map is stored and regularly updated. Wang Zhi-zhong^[4] proposed a robot path planning algorithm based on the tangent point optimization artificial potential field method in three-dimensional space, and used the tangent point method and the equidistant point method for path planning. Xiaoping

proposed three potential fields to solve the problems of global congestion and human obstacle avoidance in the actual operation of multi-robots^[5].

In a word, the research focus of domestic and foreign scholars is still on the improvement of a single path planning algorithm, but there are few studies on the combination of global path planning algorithm and local path planning algorithm. In response to this problem, this paper proposes a combined control strategy applied to the path planning of the handling robot.

II. ARTIFICIAL POTENTIAL FIELD METHOD BASED ON OBSTACLE BOUNDARY DETECTION

The artificial potential field method is one of the classic algorithms to realize the obstacle avoidance of the handling robot, but the classical artificial potential field method also exposes its inherent defects in its wide application, mainly the problem of unreachable target and local minimum problem^[12]. The details are as follows:

A. Target unreachable problem

From the above theoretical analysis, it can be seen that when there are obstacles near the target position, the robot is not affected by the repulsive force; when there is only gravity, it moves in the direction of the target. As the robot approaches the target point, it is affected by the repulsive force of nearby obstacles. It is getting bigger and bigger; at a certain position, when the gravitational force exerted by the robot is less than the repulsive force, the robot still moves toward the target, but cannot reach the target point in the end, and the planned path cannot converge, as shown in Fig. 1

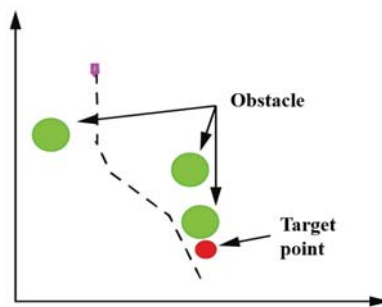


Fig. 1. Potential field change curve 1. Target unreachable phenomenon

B. Local minimum problem

In the artificial potential field method, the local minimum problem refers to that in the process of avoiding obstacles in the classical artificial potential field method, the robot is temporarily stagnant and hovering due to the influence of the position and shape of different obstacles. This phenomenon is local minima, as shown in Fig. 2.

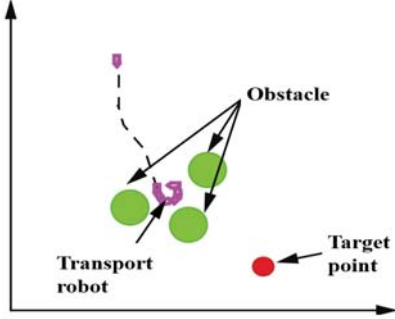


Fig. 2. Potential field change curve 2. minima phenomenon

According to the research of the classic artificial potential field method, in order to solve the problem of local minima and unreachable target when the handling robot uses the path planning, a correction factor is added, which essentially considers the relative distance between the handling robot and the target. The repulsion potential field function of is as follows:

$$U_{rep}(X) = \begin{cases} \frac{1}{2} K_{rep} \left(\frac{1}{d(X, O)} - \frac{1}{d_0} \right)^2 d^n(X, G) & d(X, O) < d_0 \\ 0 & d(X, O) \geq d_0 \end{cases} \quad (1)$$

In the formula, $d(X, G)$ represents the relative distance between the handling robot and the target position, and n represents the repulsive force field correction factor.

Calculate the direction angle and path point position of the next movement by formula (2):

$$\theta = \arctan \frac{F_{atty}(X, X_t) + F_{repy}(X, X_o) + F_{rep2y}(X, X_o)}{F_{attx}(X, X_t) + F_{repx}(X, X_o) + F_{rep2x}(X, X_o)} \quad (2)$$

$$\begin{cases} x = x + l \times \cos \theta \\ y = y + l \times \sin \theta \end{cases} \quad (3)$$

In the formula, l represents the algorithm step length, x and y indicate the position of the next waypoint.

III. POTENTIAL FIELD ANT COLONY ALGORITHM

In the potential field ant colony algorithm, the heuristic information is composed of the potential field force heuristic information and the distance heuristic information, which can be expressed as:

$$\eta_{ij}(t) = \eta_F(t) \cdot \eta_d(t) \quad (4)$$

In the formula, $\eta_d(t)$ represents distance heuristic information, and η_F represents potential field force heuristic information. Where $\eta_d(t)$ and η_F can be expressed as:

$$\eta_d(t) = \frac{1}{d(P, T)} \quad (5)$$

$$\eta_F(t) = \lambda \cdot a^{F \cdot \cos \theta} \quad (6)$$

$$\lambda = \frac{K - K_m}{K} \quad (7)$$

In the formula, $d(P, T)$ represents the distance between the current node P and the target point T , λ represents the influence coefficient of the heuristic information indicating the potential field force, K and K_m represent the maximum number of iterations and the current number of iterations in obstacle avoidance, respectively; a is a constant and $a > 1$, F represents the force of the potential situation, and θ represents the angle between the force of the potential situation and the feasible path.

IV. SIMULATION ANALYSIS AND EXPERIMENT

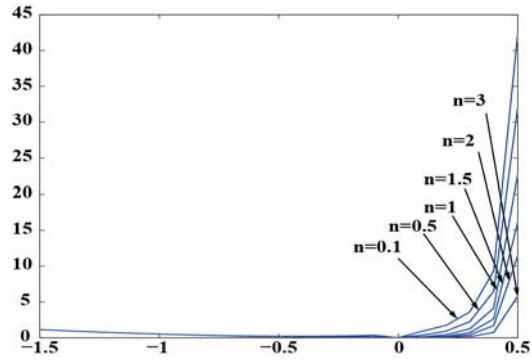


Fig. 3. Potential field change curve

If the target position is at the coordinate origin (0, 0), the obstacle position is (0.6, 0), and the current path point of the handling robot (-1.5, 0) is moving toward the target point along the positive direction of the x axis. Fig. 3 shows that

when $K_{rep} = K_{att} = 1$, when n is 0.1, 0.5, 1, 1.5, 2, 3, the total potential field change experienced by the handling robot. It can be seen from Fig. 3 that when $n > 1$, there is only a global minimum. When $n < 1$, the potential field curve is deformed, and there is a local minimum in the process of approaching the target point. The smaller the n , the larger the total potential field curve deformation of the handling robot when it approaches the target point, and there is a local minimum, and there is a local minimum. Therefore, in order to ensure path convergence and avoid local minima, the correction factor n should be greater than 1. However, if the value of n is too large, it will increase the amount of calculation of the algorithm, which is not conducive to the effective execution of the algorithm, and reduces the

efficiency of the algorithm. Therefore, $n=2$ can be selected, and the total potential field function can be expressed as:

$$U_{art} = \begin{cases} \frac{1}{2}K_{rep}(\frac{1}{d(X,O)} - \frac{1}{d_0})^2 d^2(X,G) + \frac{1}{2}K_{att}d^2(X,X_t) & d(X,O) < d_0 \\ \frac{1}{2}K_{att}d^2(X,X_t) & d(X,O) \geq d_0 \end{cases} \quad (8)$$

When $K_{rep} = K_{att} = 1$, $d_0 = 2$, $n=2$, as shown in Figure 4, the target position is at the origin of the coordinates (0, 0), the obstacle position is (0.6, 0), and the current path point of the handling robot (-1.5, 0), it can be seen from Fig. 4 that in the potential field environment where the handling robot is moving towards the target point, the total potential field of the handling robot at the target point is the smallest, that is, the potential field is the global minimum value, and there is no local minimum point, it can be proved that the potential field function constructed with this parameter is reasonable.

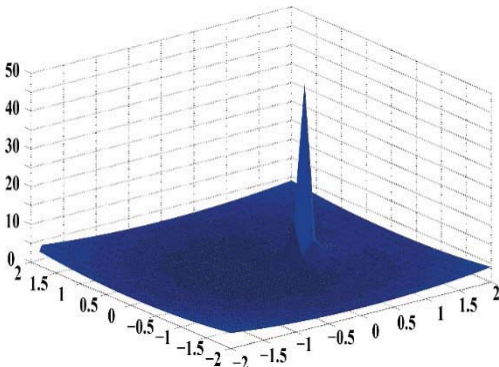


Fig. 4 Potential field change curve 4. The total potential field change surface

Set the relevant parameters $K_{rep} = K_{att} = 1$, $d_0 = 2$, $n = 2$, and use the improved artificial potential field method to simulate the path planning. As shown in Fig. 5 and Fig. 6, assuming that the starting point coordinates are (0,0), the target point coordinates are (10,10), and the obstacle coordinates are (1.5,1.5), (3,3), (4, 4.5), (3,6), (3,3), (4.5,4), (5.5,5.5), (6,2), (8,7), (8,8), (9,8), under the same conditions as above, the conventional artificial potential field algorithm and the improved artificial potential field method with correction factor added to solve the problem of unreachable target are used to simulate obstacle avoidance. From the analysis of Fig. 5 and Fig. 6, it can be seen that in a two-dimensional space, if there is an obstacle near the target point, using the conventional artificial potential field method, the planned path will cross the obstacle near the target point (Fig. 5). The path is actually infeasible. Using an improved algorithm that adds a correction factor, the planned path can effectively avoid obstacles near the target point (Fig. 6), so the improved algorithm is effective.

V. SIMULATION ANALYSIS OF POTENTIAL FIELD ANT COLONY ALGORITHM

According to the steps of the improved algorithm, a simulation analysis is performed. The setting of the ideal algorithm parameters is obtained through a large number of experiments. Set the gravitational gain system $k = 20$,

repulsion gain coefficient $\eta = 10$, and the initial step length $l^* = 0.1m$; the population number setting cannot reduce the optimization ability of the algorithm, set the number of H groups $m=50$, the relative importance of pheromone and heuristic information is set to $\alpha = 1$, $\beta = 10$ (can not reduce the performance of pheromone and heuristic information), the evaporation rate is set to $\rho = 0.4$, and the initial

pheromone value is $\tau_0 = 80$. According to the specific size of the handling robot $6.28m \times 2.22m$, according to the grid method environment modeling theory, set the grid side length to 6.3m, assuming that the offline map is a 20×20 binarized grid map, the simulation result is shown in Fig. 7. The red solid line represents the shortest path, and the blue solid line represents other paths planned by the two algorithms. On the 20×20 grid model, the potential field ant colony algorithm calculates fewer paths.

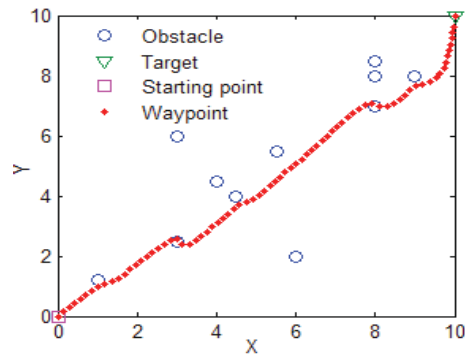


Fig. 5. Path planning simulation trajectory of conventional APF algorithm

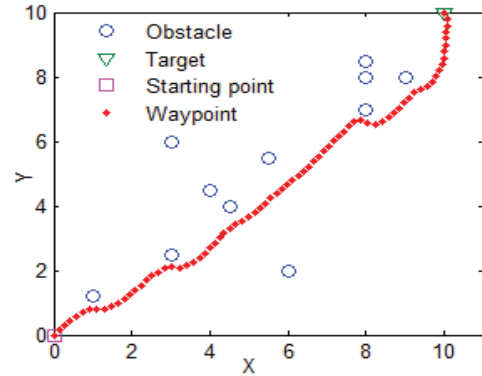
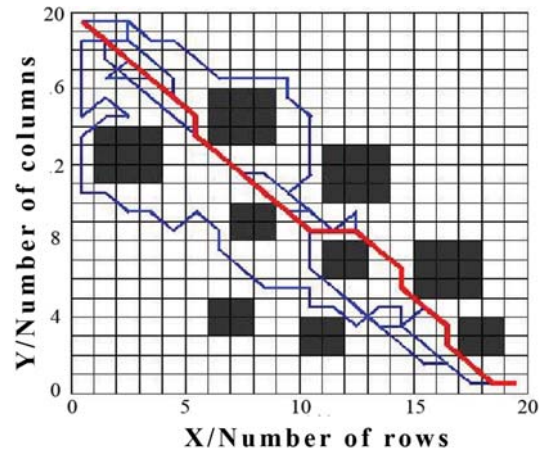
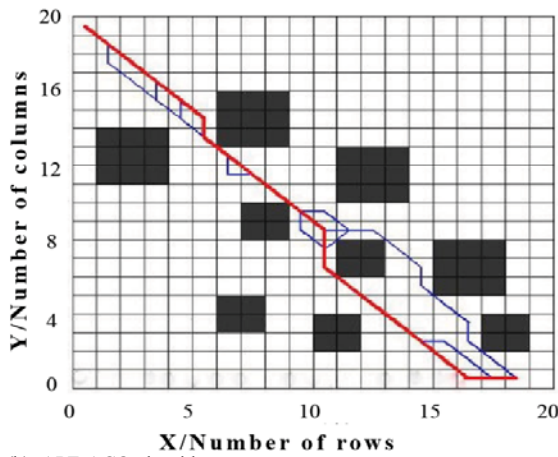


Fig. 6. Improved APF algorithm path planning simulation trajectory



(a) Conventional ACO algorithm



(b) APF-ACO algorithm

Fig.7. ACO algorithm and APF-ACO algorithm path planning simulation results

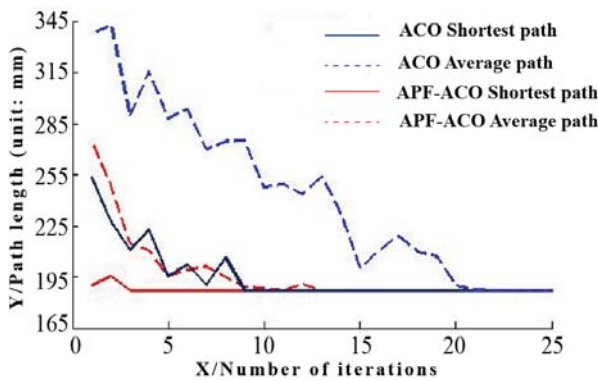


Fig.8. Comparison of APF-ACO algorithm and conventional ACO algorithm

As shown in Fig. 8, the red and blue dashed lines respectively represent the average path length of the improved potential field ant colony algorithm and the average path length of the conventional ant colony algorithm during the obstacle avoidance simulation process. The red solid line and the blue solid line respectively represent the potential field Ant colony algorithm path length and conventional ant colony algorithm path length. As shown in Fig. 8, the convergence curve of the potential field ant colony algorithm is significantly faster than the conventional ant colony algorithm.

VI. CONCLUSION

The path planning control strategy of the handling robot is the key task of the navigation control of the handling robot. Aiming at the problems of the traditional artificial potential field method and the ant colony algorithm, this

paper proposes a combined control combining the improved artificial potential field method and the improved ant colony algorithm Strategy. In local path planning, the traditional artificial potential field method is improved by the method of increasing the correction factor. The improved artificial potential field method is better than the traditional algorithm, and it can effectively solve the ubiquitous target unreachable and local minimum. Inherent defects are feasible. Under this premise, the traditional ant colony algorithm and the improved artificial potential field method are combined to form the potential field ant colony algorithm. In the same grid map environment, the potential field ant colony algorithm is compared with the traditional ant colony algorithm. The improved potential field ant colony algorithm reduces the number of iterations of the shortest path by 67%, the maximum number of iterations is reduced by 38.1%, and the shortest path length and average path length are reduced by 2% and 1.5% respectively. Therefore, the trajectory planning effect of the improved algorithm is better.

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